

Open Data Portal Port Management



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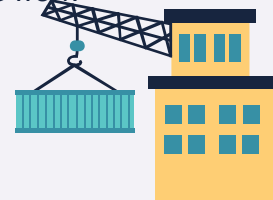
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01

The Team



André Alves



Architect

Pedro Rei



DB master

Francisco Pinto



Project Manager

José Moreira



Advisor

Diogo Costa



Product Manager

Bruno Tavares



Devops

Paulo Pintor

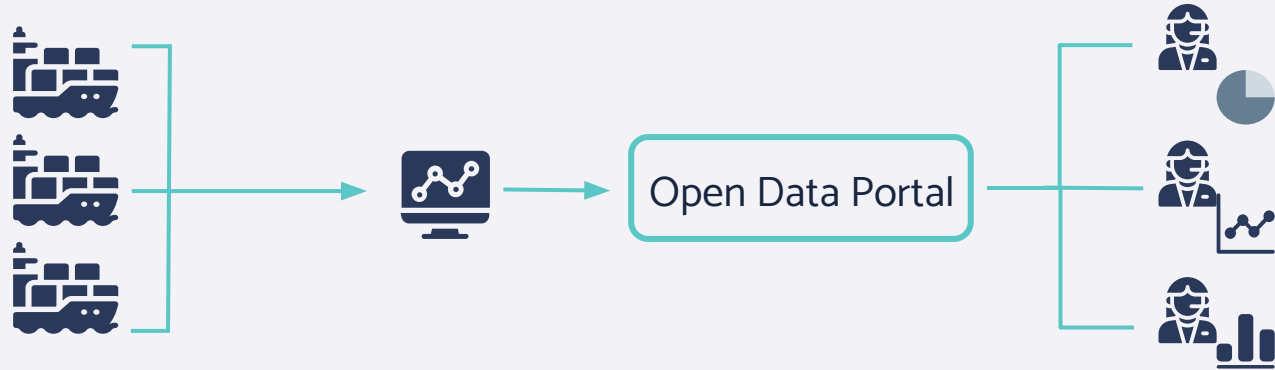


Advisor

01

Context

The need for a single portal for port management data analytics with data provenance.



01

Problems and Motivation

The current system used by the Port of Sines lacks clear metadata and **data provenance** tracking, making it challenging for users to determine the origin, transformation history, and reliability of the data. Additionally, the presence of **missing or incorrect data** further complicates data analysis and interpretation.

Our goal is to develop a platform that not only enables users to **visualize and analyze** detailed information but also provides full transparency regarding the origins and processing of the data. At the same time, the platform will address and correct issues related to missing and inaccurate data present in the original system.



01

State of the Art



OpenPort ¹



Softship ²



Piveau ³



Open Data Portal ⁴

Data Storage	✓	✓	✓	✓
Analytics	✓	✓	✓	✓
Maritime Data	✓	✓	✗	✓
Data Provenance	✗	✗	✓	✓
Data completeness with ML model	✗	✗	✗	✓
Portuguese	✓	✗	✗	✓

1-<https://openport.com.br/>

2-<https://www.softship.com/>

3-<https://www.piveau.de/en/>

4-<https://pi-open-data-portal.github.io/>

02



Functional Requirements

Administrator should be able to:

- Manage User accounts
- View User interaction with the system
- Manage Data quality

User should be able to:

- Login/register
- See the data and analytics
- Interact with and filter the evaluation metrics
- Download and export data in different formats
- Report data errors
- See provenance of data
- Track origin of data



02



Non-functional Requirements



Performance



Data Protection
and Governance



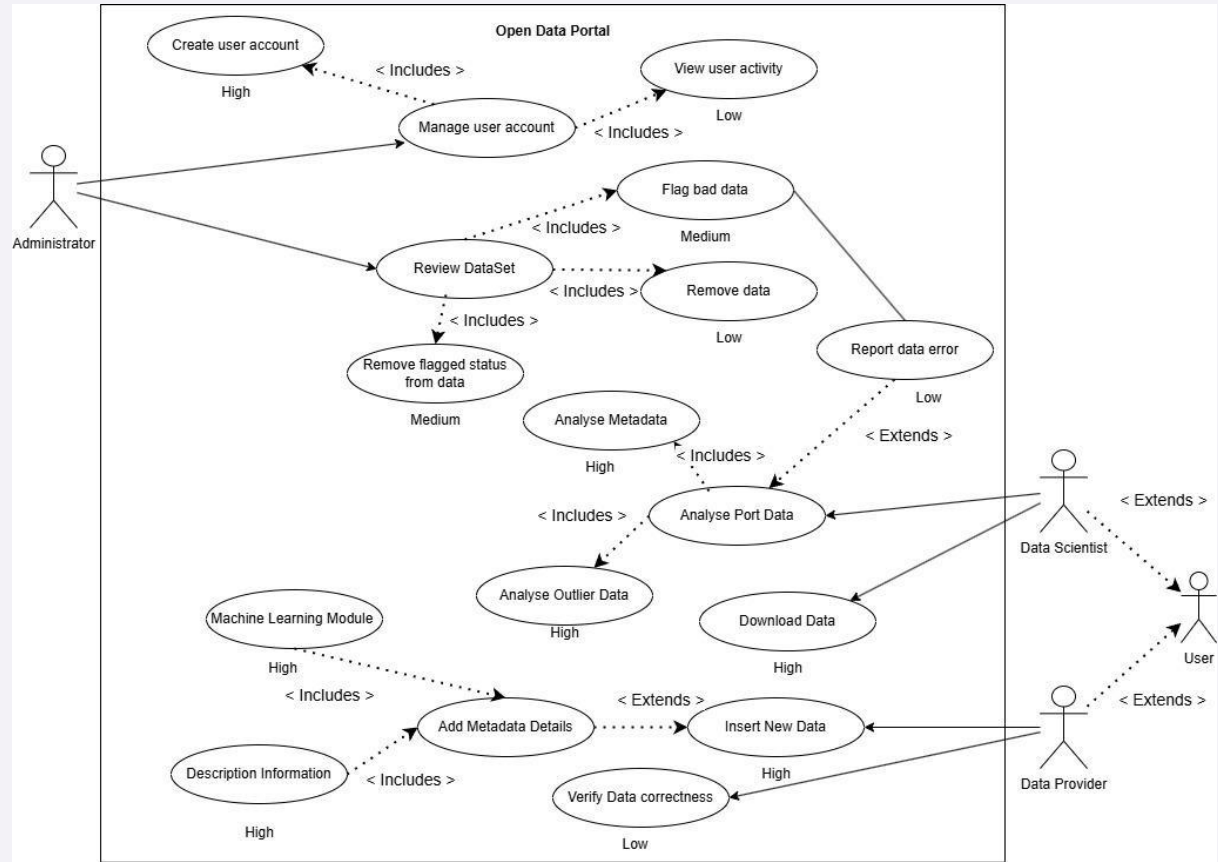
Usability



Maintainability

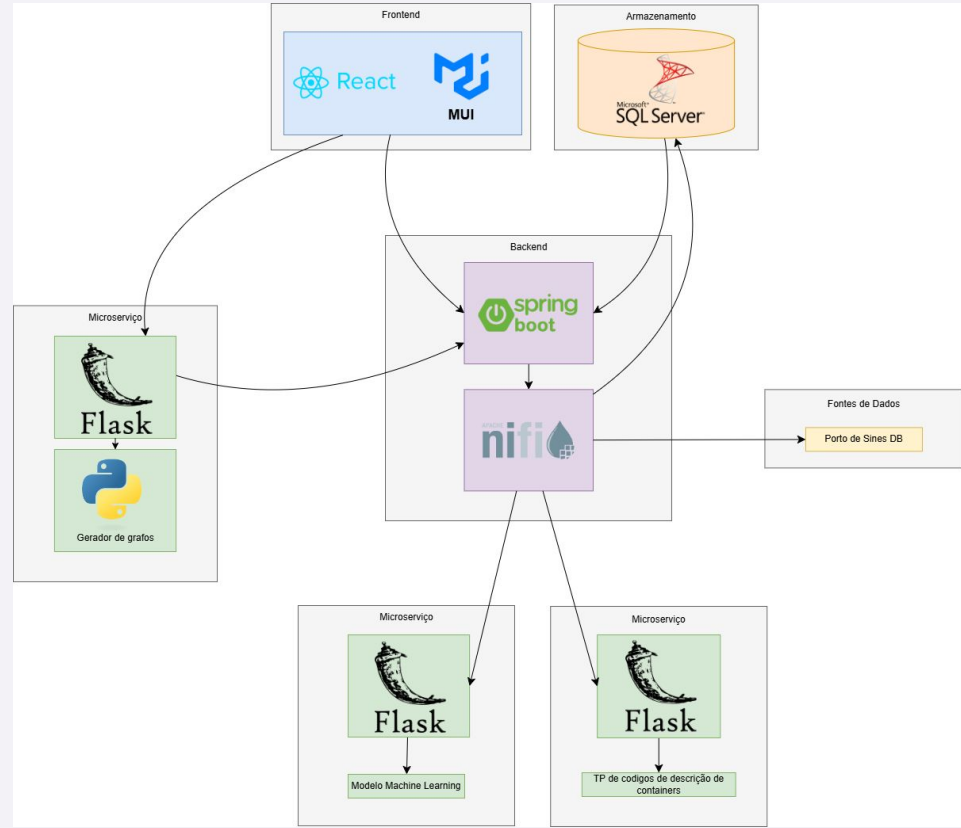


02 Use Cases



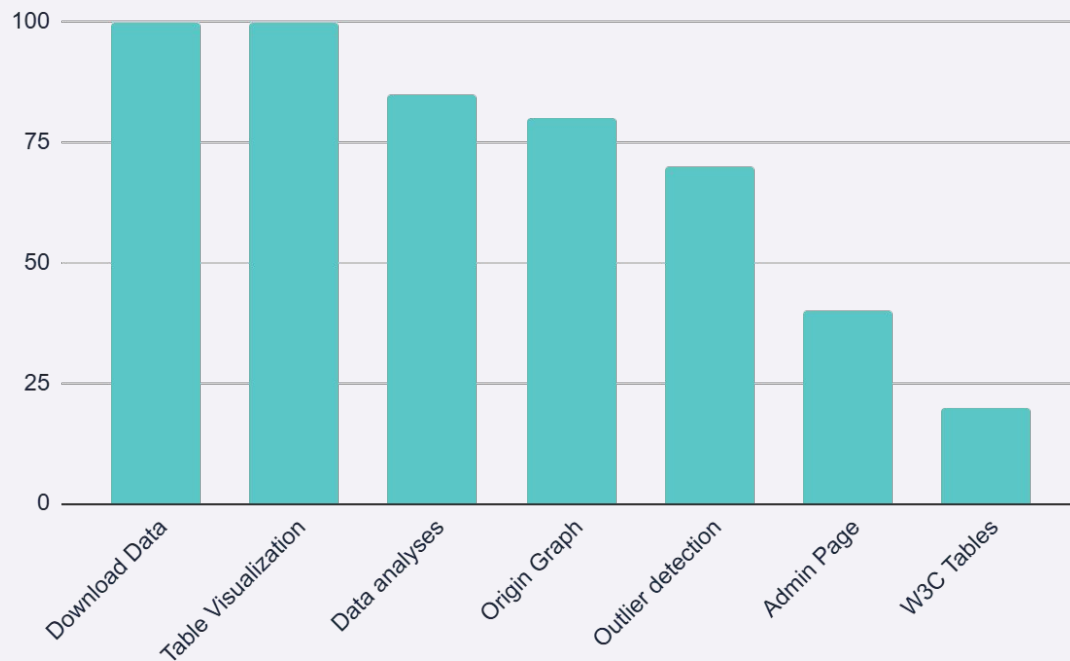
03

System Diagram



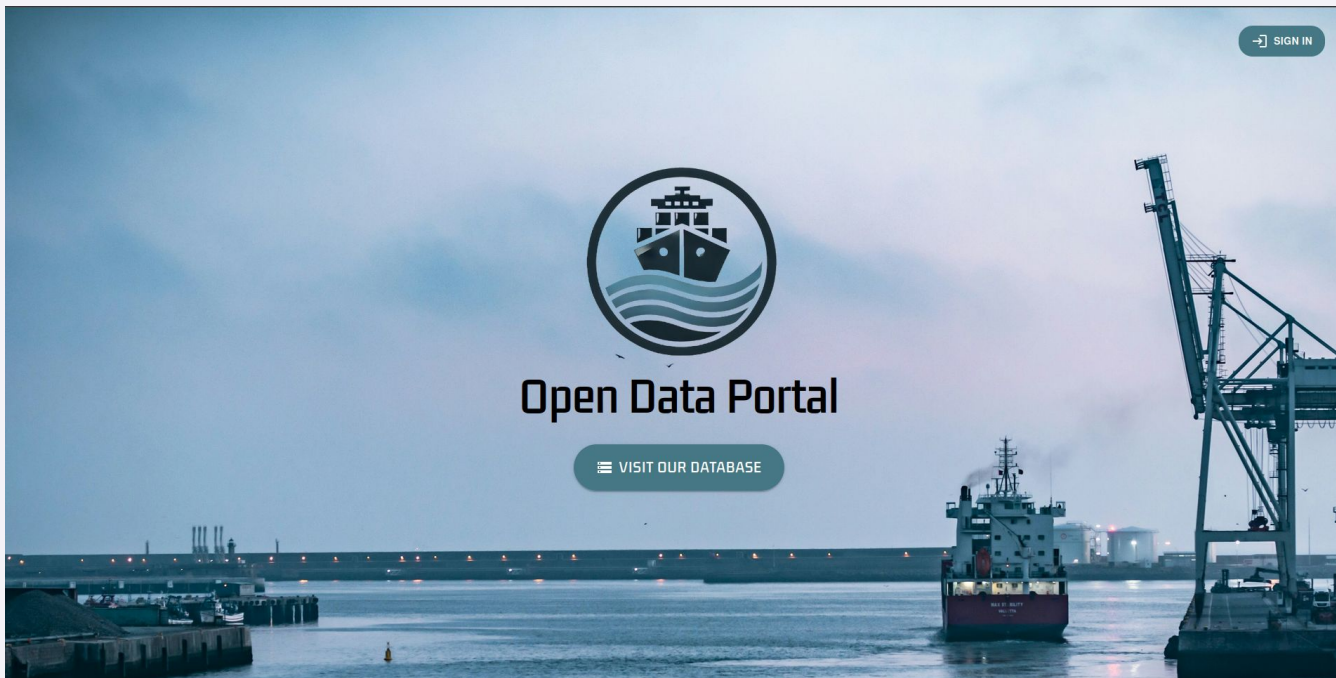
03

Tasks



04

Frontend



04

Frontend



Example Port

CONTAINER
DETAILSLOGISTICS DATA
DASHBOARDCODES
INFORMATION

Filter by Prov2

Search by Container Plate

Search by Code

Start Date

dd/mm/aaaa

End Date

dd/mm/aaaa

COLUMNS

DOWNLOAD

ID	Container Plate	Cargo Description	Weight (kg)	NST 3P	Prov2	Actions
1	FCIU4465304	GLYCOLINE, XTERN RUMIANTES 1.5%	17 860	047	T1	👁 📍
3	MSMU4146181	NEW PNEUMATIC TYRES, OF RUBBER - OF A KIND USED O NEW PNEUMATIC TYRES, OF RUBBER - OF A KIND USED O NEW PNEUMATIC TYRES, OF RUBBER - OF A KIND USED O NEW PNEUMATIC TYRES, OF RUBBER - OF A KIND USED O NEW PNEUMATIC TYRES, OF RUBBER - OF A KIND USED O NEW PNEUMATIC TYRES, OF RUBBER - OF A KIND USED O	14,94	052	T2	👁 📍
6	TCLU7074739	.GRAPHITE ELECTRODE 650 X 2700 MM WITH CONNECTING PIN 355 X 559 MM, 4TPI	166 626	118	T3	👁 📍
7	MSDU5124920	1 SET OF BRAKE PADS FOR DISK BRAKE 99135194982NB DE NCG 87083091 1 SET: WIPER BLADES 9J1955425DYNB PL NCG 85129090 AIR GUIDE MATT BLACK 9Y0117335ACOK1NB DE J CG 87089199 AS ROOFLINER 981555091166F2NB PL N CG 87082990 AS ROOFLINER 981555091166F2NB PL N CG 87082990 AS ROOFLINER 981555091166F2NB PL N CG 87082990 AS SUPPORT BEARING FA,997 C4 9973430150	0,85	118	T4	👁 📍
9	MSDU7940440	185/65R15 92T XL VC7 HS:40111000 225/50R18 99V XL PROC SSR * HS:40111000 225/50R18 99V XL PROC SSR * TIRES HS-CODE:4011 1000 HS:40111000 225/60R17 103T ALT ARC 12 HS:40111000 235/50R18 97H PROCTX SSR MOE HS:40111000 235/60R17 106T ALT ARC 12 HS:40111000 235/60R17 106T ALT ARC 12 HS:40111000 235/60R18 107V XL CCLXSP LR HS:40111000 245/45ZR18 100Y XL	14,16	052	T5	👁 📍
10	MEDU4195201	BOARD BOOK BOOKS BOTTOM SHEET CHILDREN BOOKS CUTTING MACHINE MOTORCYCLE SPARE PARTS	1953,65	102	T6	👁 📍

Rows per page: 25

1-25 of 96086

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04

Frontend

The screenshot displays the 'Porto de Aveiro' web application. A modal window titled 'Cargo Container Details - FCIU4465304' is open, showing detailed information for a specific container. The background interface includes a sidebar with navigation options: SHIP DETAILS, CONTAINER DETAILS, PROVENANCE STATISTICS, EMBARCATION PORTS, and PRODUCT CATEGORIES. The main area shows a table of containers with columns for ID, Container Plate, and Ca. A 'Filter by Prov2' dropdown and a 'DOWNLOAD' button are also visible. The modal contains three sections: Container Information, Cargo Information, and Shipping Information.

Container Information

Container ID	Container Plate	State
1	FCIU4465304	Cheio
ISO Container	ISO Registry	Container Tare
22G1	22G1	2200

Cargo Information

Cargo Description
GLYCOLINE, XTERN RUMIANTES 1.5%

Weight (kg)	Departure Weight	Packages Quantity
17 860	17 860	20
Broken Packages	Harmonized Code	
20	29053200	

Shipping Information

Movement Date	Embarkation Port	Disembarkation Port
2023-06-15	PTSIE	CLSAI
Transhipment		
S		

[CLOSE DETAILS](#)

04

Frontend

The screenshot shows a web application interface for 'Example Port'. On the left is a sidebar with navigation links: 'CONTAINER DETAILS', 'LOGISTICS DATA DASHBOARD', and 'CODES INFORMATION'. The main area displays a table of container data with columns: ID, Container Plate, Cargo Description, Weight (kg), NST 3P, Prov2, and Actions. A 'Download Data' modal is open in the center, allowing users to select columns and format for download.

Download Data Modal:

Select the columns and format for the download

- ☒ ID
- ☒ Container Plate
- ☒ Cargo Description
- ☐ Message
- ☐ Movement Date
- ☐ Embarkation Port
- ☐ Disembarkation Port
- ☐ Transhipment
- ☐ ISO Container
- ☐ ISO Container Registry
- ☐ Container Tare
- ☐ Container State
- ☐ Harmonized Code
- ☒ Weight (kg)
- ☐ Broken Packages Qty
- ☐ Packages Qty
- ☐ Departure Weight
- ☒ NST 3P
- ☐ NST 2P
- ☒ Prov2
- ☐ Prov

Select the format for the download:

Format:

[CANCEL](#) [DOWNLOAD](#)

04

Frontend



Example Port

CONTAINER
DETAILS

LOGISTICS DATA
DASHBOARD

**CODES
INFORMATION**

Search by ID

Search by Code

Start Date



End Date



ID	Code	Creation Date
000143fd-b039-42d0-b2eb-b0f389530686	4765	17/10/2020
00031b59-f785-4976-86f9-919ba9661cd1	7LH9	17/10/2020
00049acd-0114-413b-8402-08789836c62e	2771	17/10/2020
0004d91-d224-4504-baee-4230a3370aea	1366	17/10/2020
0007e61e-572d-49c0-bd3d-97dc2ea1edba	2DKD	15/11/2023
000b3dde-6411-4e24-a86e-5064bb2b1028	2CH2	17/10/2020
00117c03-b0d0-4714-b36b-45290476ebe5	1446	17/10/2020
0019e6e5-777c-4f1a-a1c7-4184deda5d82	28GP	17/10/2020
0030f1ba-afid-4cb6-af58-23687d62e0b8	7DB3	17/10/2020
00339704-8c88-4246-a9a0-07f6ba4619f1	39H6	17/10/2020

Rows per page: 25 ▾ 1-25 of 10582 < >



04

Frontend

LOGISTICS DATA
DASHBOARD

CODES
INFORMATION

Port Pairs Analysis

Connections between embarkation and disembarkation ports

Filters

Start Date



End Date



Message Type



Embarkation Ports



Disembarkation Ports



☐ Is Transshipment

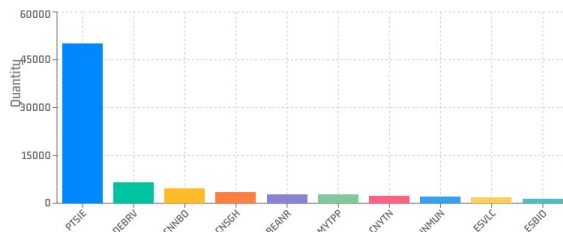
X CLEAR

APPLY FILTERS

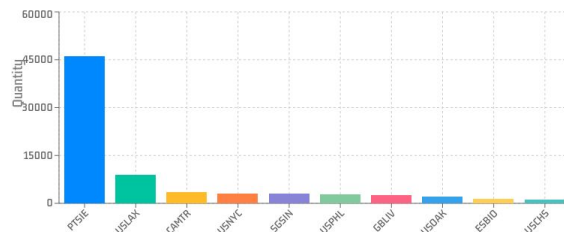
GRÁFICOS

DATA TABLE

Top Embarkation Ports

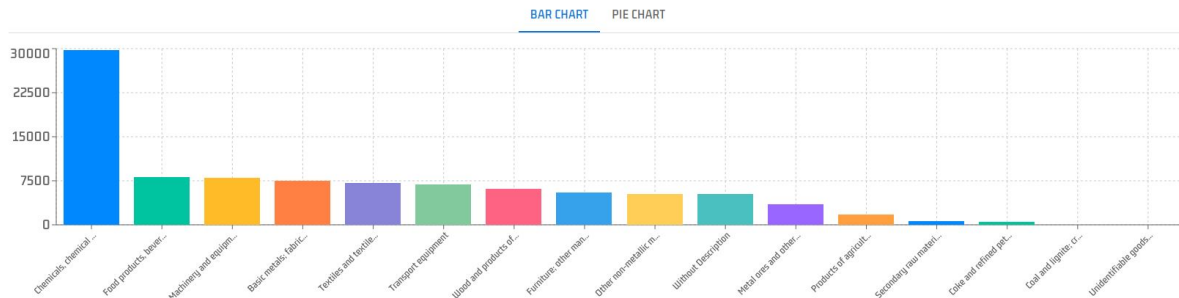


Top Disembarkation Ports



Product Classification Analysis

Distribution of Products by Category

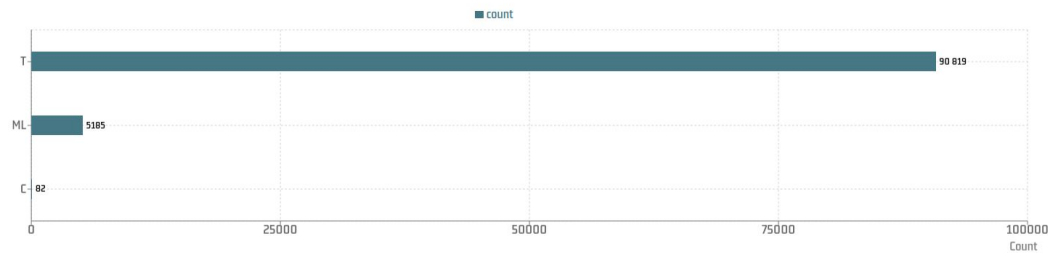


Product Category	Count	Percentage
Chemicals, chemical products, and man-made fibres; rubber and plastic products; nuclear fuel	29 794	31.01%
Food products, beverages and tobacco	8 087	8.42%
Machinery and equipment n.e.c.; office machinery and computers; electrical machinery and apparatus n.e.c.; radio, television and communication equipment and apparatus; medical, precision and optical instruments; watches and clocks	8 011	8.34%
Basic metals; fabricated metal products, except machinery and equipment	7 545	7.85%

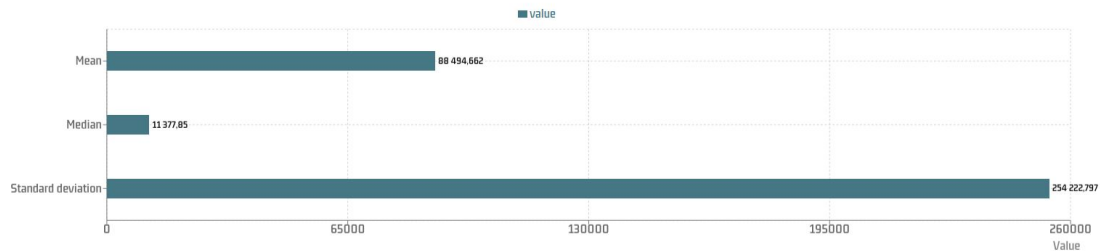
04

Frontend

Provenance Statistics



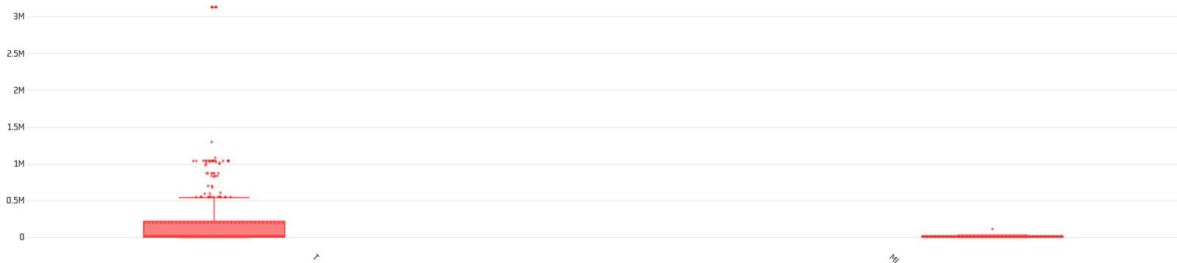
Weight Statistics



04

Frontend

Weight Distribution by Provenance



Sample size: 500 records from 2 provenance categories (categories with fewer than 5 records were excluded)

Outlier Records

Below are records with weights that fall outside the typical range for their provenance category (defined as values beyond $1.5 \times \text{IQR}$ from the first and third quartiles).

ID	Weight	Provenance	Movement Date
51555	118020	ML2617	2023-06-01
10852	3131916	T8398	2023-06-01
12665	3131916	T10071	2023-06-01
86115	3131916	T77995	2023-06-01
17930	3131916	T14915	2023-06-01
14681	3131916	T11917	2023-06-02
72969	3131916	T65832	2023-06-02
63287	1300260	T56884	2023-06-02
16031	1087700	T13160	2023-06-01
65083	1044438	T58552	2023-06-02

05

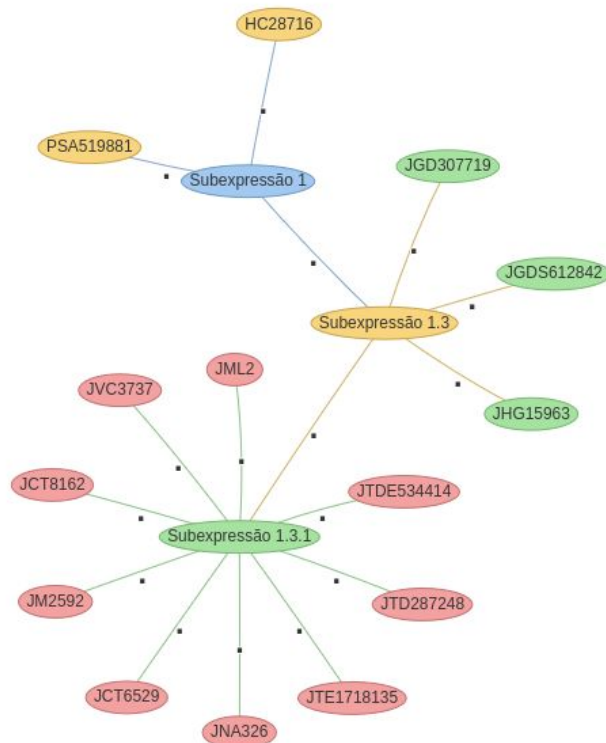
Backend

```
SELECT Container_Plate
FROM case_study
GROUP BY Container_Plate
```

Container_plate	Provenance
ALPU2220 618	⚡(E_jultosPSA_trans equip_movement:PSA2332980 . ((E_julgn_vessel_call:JVC4891 . E_julgn_notice_of_arrival:JNA2483 . E_julcm_manifest:JM2447 . E_julcm_transport_document:JTD458301 . E_julcm_transport_document_equipment:JTDE491887 . E_julmd_container_type:JCT8047 . E_julmd_transport_equipment:JTE3531291 . E_julmd_container_type:JCT7627 . jul_md_lov:JML2) . E_julcm_goods_detail:JGD485249 . E_julcm_goods_detail_split:JGDS1369476 . E_julmd_harmonized_goods:JHG7520) . HC_NST2007:HC23623 + E_jultosPSA_trans equip_movement:PSA1356210 . ((E_julgn_vessel_call:JVC4398 . E_julgn_notice_of_arrival:JNA2406 . E_julcm_manifest:JM2154 . E_julcm_transport_document:JTD208216 . E_julcm_transport_document_equipment:JTDE1044806 . E_julmd_container_type:JCT8047 . E_julmd_transport_equipment:JTE3531291 . E_julmd_container_type:JCT7627 . jul_md_lov:JML2) . E_julcm_goods_detail:JGD116342 . E_julcm_goods_detail_split:JGDS1228255 . E_julmd_harmonized_goods:JHG7520) . HC_NST2007:HC23623)

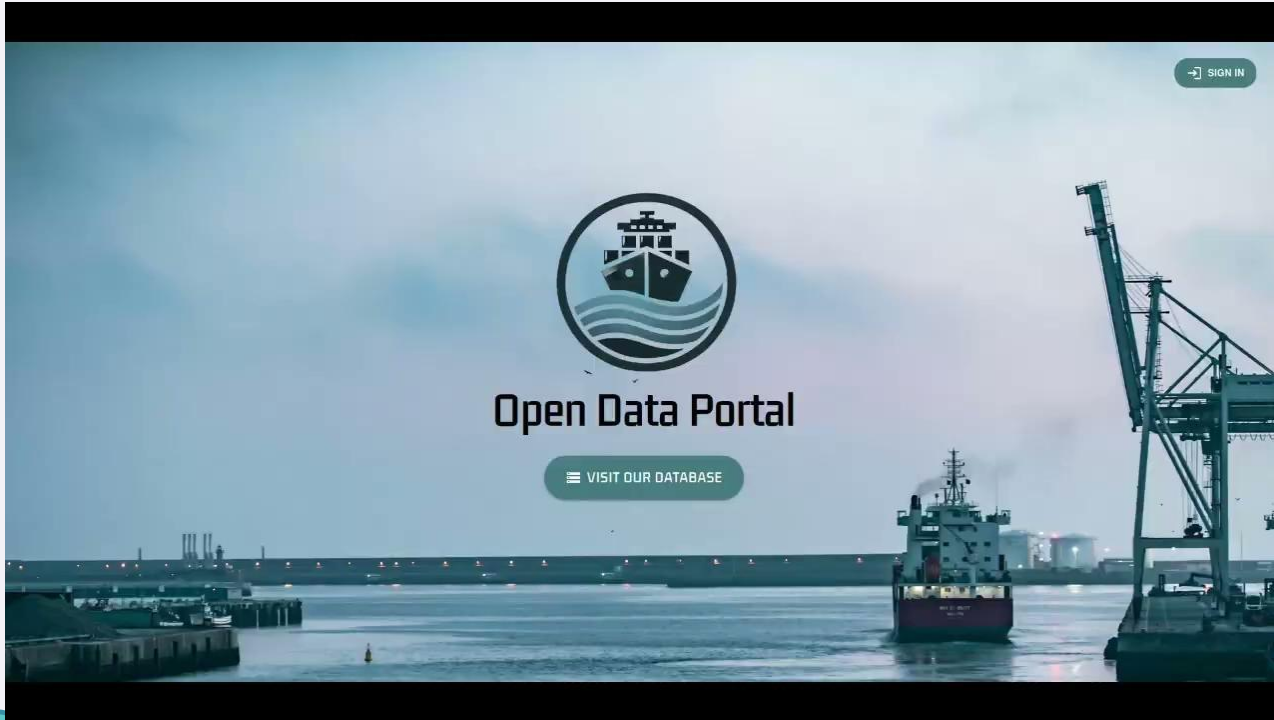
05

Backend



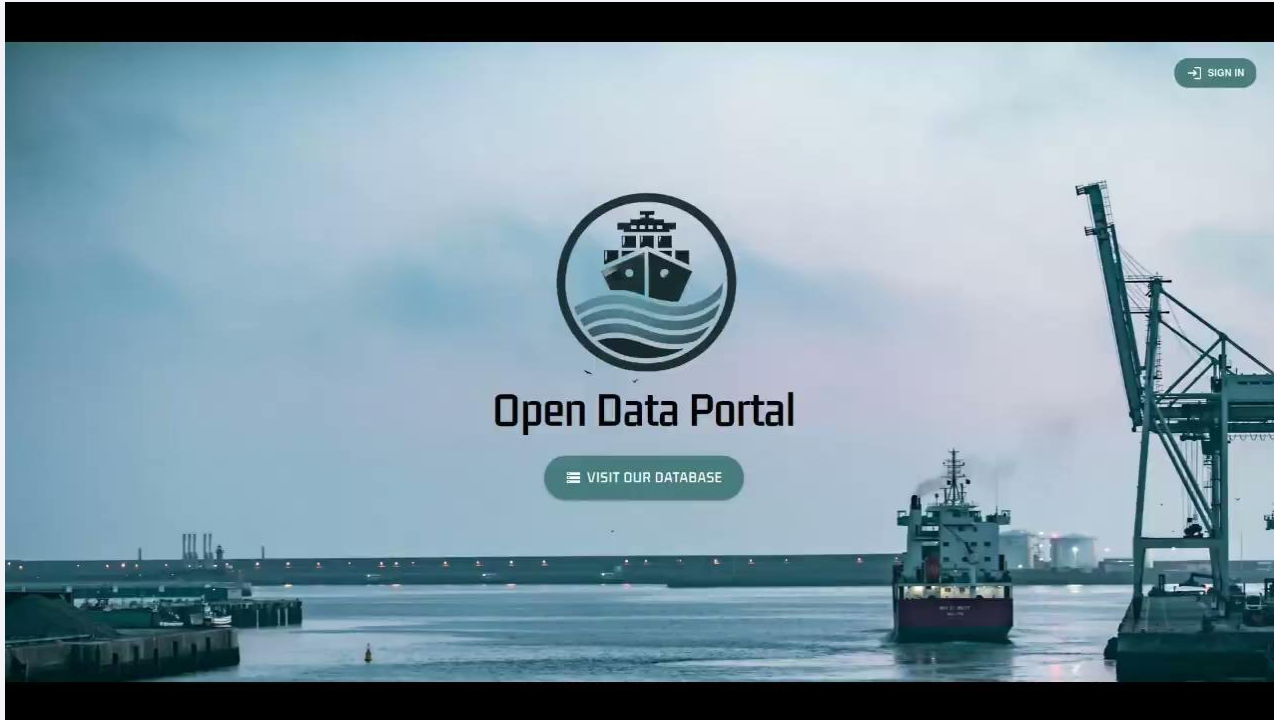
06

Demo:Data analysis



06

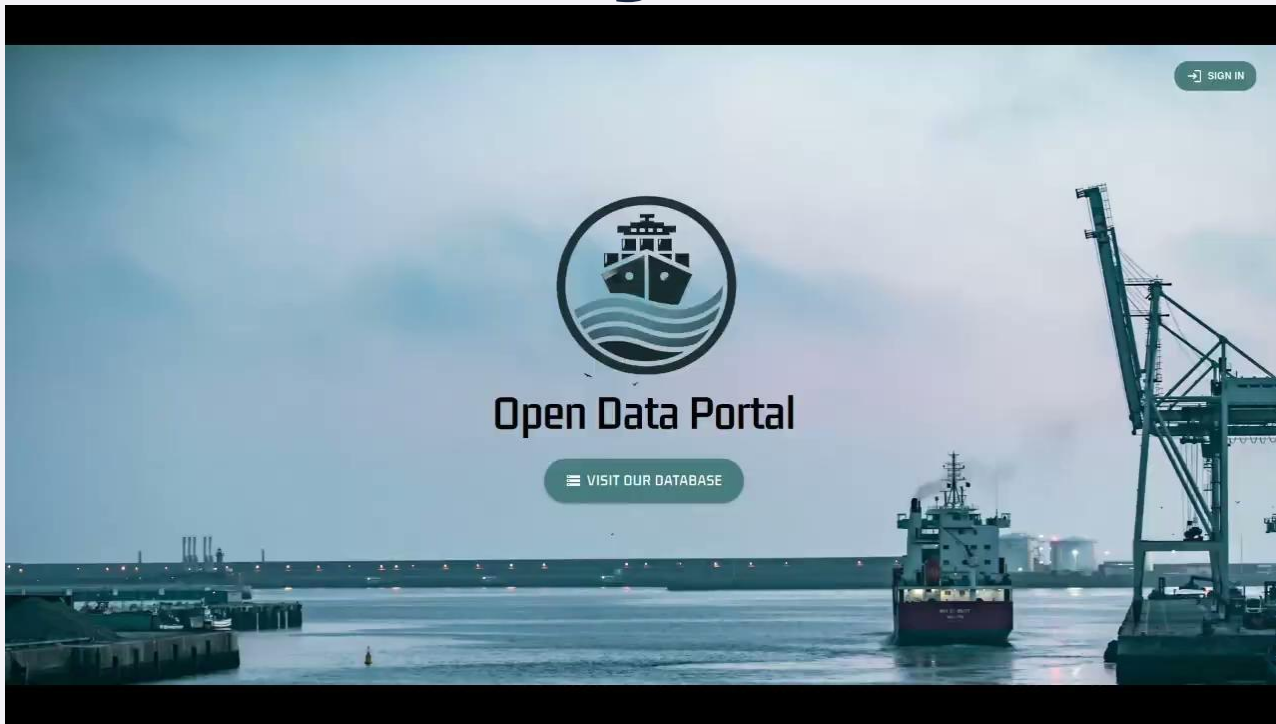
Demo:Download data



06



Demo:View data origin



07



Future Work

- Add normalization to Graphs of origin to improve visualization for complex queries
- Add caching to improve performance
- Add multiple outlier functions options
- W3C tables
- Development of case study 2
- Track system metrics



07



Calendar

Week

Steps	9	10	11	12	13	14	15	16	17
Origin graph development									
Dynamic Outlier detection									
W3C tables Analyses									
Case study 2									
Admin Metrics									
User authorization									
ETL pipeline									
Deployment									
Final Report									



Thank you



Open Data Portal
Port Management



<https://pi-open-data-portal.github.io/>

